

Notes: Auffhammer and Kellogg (AER, 2011)

Clearing the Air? The Effects of Gasoline Content Regulation on Air Quality

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1 Big picture

- **Question.** Do gasoline content regulations reduce ground-level ozone pollution, and does the answer depend on how flexibly refiners are allowed to comply?
- **Setting.** The paper studies US gasoline content rules intended to reduce ozone precursors from mobile sources: federal Reid vapor pressure (RVP) limits, federal reformulated gasoline (RFG), and California Air Resources Board reformulated gasoline (CARB).
- **Core empirical variation.** The regulations are implemented at different times and places. Some counties receive stricter vapor-pressure rules, some adopt RFG, and all California counties adopt CARB in 1996.
- **Main finding.** Federal RVP and RFG rules do not measurably reduce ozone. CARB substantially reduces ozone, especially in the severely polluted Los Angeles–San Diego area.
- **Mechanism.** The federal standards target total VOC mass or evaporation but allow refiners to choose which VOCs to remove. Refiners respond by removing butane, a cheap-to-remove compound that is not very ozone-reactive. CARB instead restricts highly reactive VOC classes such as olefins and aromatics, so refiners cannot satisfy the rule merely by reducing butane.
- **Economic lesson.** Regulatory flexibility lowers compliance costs only if the regulated margin maps well into environmental damages. If cheap compliance abates the wrong pollutant component, flexible standards may create costs without meaningful benefits.

2 Incremental contribution of the paper

- **Measures benefits, not only costs.** Earlier work emphasized that gasoline content rules raise prices and segment gasoline markets. This paper directly asks whether those costly rules deliver the intended ozone reductions.
- **Compares flexible and targeted standards in the same policy domain.** Federal RVP/RFG and California CARB all regulate gasoline content, but differ sharply in regulatory flexibility and chemical targeting. This creates a clean comparison of design choices.
- **Connects environmental economics with atmospheric chemistry.** The paper does not treat VOC reductions as homogeneous. It emphasizes that ozone formation depends on VOC/NO_x chemistry, VOC reactivity, local conditions, and the specific compounds removed.
- **Uses rich spatial-temporal policy variation.** The analysis combines monitor-day ozone data, detailed gasoline-rule timing, county-level regulatory status, and weather controls from 1989–2003.
- **Employs two complementary designs.** Difference-in-differences estimates capture broad panel changes; monitor-level time-series regression discontinuity estimates use sharp regulatory start dates and allow flexible monitor-specific trends.
- **Identifies behavioral response as central.** The paper shows that a regulation’s success depends not only on its stated target, but also on how firms choose the least-cost compliance technology.

- **Policy takeaway beyond gasoline.** The results speak to a general problem in flexible environmental regulation: if marginal damages differ by location, time, or pollutant type, allowing uncontrolled flexibility can reduce environmental benefits.

3 Regulatory and chemical background

3.1 Ozone formation and why VOC details matter

Ground-level ozone is not emitted directly. It forms when volatile organic compounds (VOCs) react with oxides of nitrogen (NO_x) in the presence of heat and sunlight. This has three implications for gasoline regulation.

- **Seasonality.** Ozone formation is strongest during warm, sunny months. Therefore, RVP, RFG, and CARB VOC components are especially relevant in summer.
- **VOC-limited versus NO_x -limited areas.** In some areas, reducing VOC emissions lowers ozone. In other areas, ozone is constrained by NO_x , so cutting VOCs may do little.
- **VOCs are heterogeneous.** A pound of one VOC is not equivalent to a pound of another. Some compounds are much more reactive in forming ozone. A rule that reduces cheap but weakly reactive VOCs may show little ozone benefit.

Interpretation: This chemistry is the reason a simple “VOC reduction” rule can fail. The economically cheapest VOC to remove is not necessarily the environmentally important one.

3.2 Federal RVP regulation

RVP regulation limits gasoline’s vapor pressure during the summer. Lower RVP reduces evaporative VOC emissions. Phase I began in 1989 with state-level standards. Phase II began in 1992 and required many ozone nonattainment counties, especially in southern states, to adopt tighter summer limits such as 7.8 psi or lower.

Economic message: RVP is a flexible performance-style regulation. It restricts overall vapor pressure but does not force refiners to reduce particular high-reactivity VOCs.

3.3 Federal RFG regulation

Federal reformulated gasoline began in 1995 in severe ozone nonattainment areas and in opt-in areas. RFG included requirements such as lower benzene, oxygen content, reductions in VOCs and toxic air pollutants, and later NO_x reductions under Phase II.

Interpretation: RFG is stricter than basic RVP, but it still allows refiners to achieve VOC reductions in ways that may not target the most ozone-reactive compounds.

3.4 California CARB reformulated gasoline

California’s CARB gasoline began statewide in March 1996. It imposed a 7.0 psi RVP limit and placed direct content limits on olefins and aromatic hydrocarbons, which are highly reactive in forming ozone. It also imposed sulfur reductions that improve on-board NO_x controls.

Takeaway: CARB is less flexible but more chemically targeted. It is designed to force removal of the compounds most relevant for ozone formation, rather than merely reducing total VOC mass cheaply.

4 Data

4.1 Air quality monitor data

The paper uses hourly ozone readings from the EPA Air Quality System for 1989–2003. The authors aggregate these readings to monitor-day outcomes:

- **Daily maximum ozone concentration.**
- **Daily maximum eight-hour average ozone concentration.**

They apply EPA-style data-quality restrictions: a valid monitor-day needs sufficient hourly observations, and a monitor-year is dropped if too many summer days are missing. The core sample contains more than one million monitor-day observations across hundreds of counties.

4.2 Weather data

Because ozone formation depends heavily on temperature and sunlight, the paper matches each ozone monitor to nearby weather stations and constructs daily weather controls. These controls include maximum and minimum temperature, precipitation, and flexible interactions with seasonality and day of week.

Interpretation: Weather controls are not merely precision improvements. Figures 3 and 4 show that ozone and temperature comove strongly, so ignoring weather could confound treatment effects with hot summers.

4.3 Regulation data

For each county and day, the authors code whether the county is subject to:

- RVP Phase I standards weaker than the 9.0 psi baseline, such as 9.5 or 10.5 psi.
- RVP Phase II standards of 7.8 psi or lower.
- Federal RFG.
- CARB gasoline.

The baseline category is summer RVP of 9.0 psi. All reported regulatory effects are interpreted relative to this omitted baseline.

5 Empirical strategy

5.1 Difference-in-differences design

The main difference-in-differences model is estimated at the monitor-day level during the summer ozone season:

$$\ln(y_{it}) = \alpha \cdot Treat_{ct} + \mu_i + \eta_{ry} + \varepsilon_{it},$$

where y_{it} is ozone at monitor i on day t , $Treat_{ct}$ is a vector of county-by-date regulatory treatments, μ_i are monitor fixed effects, and η_{ry} are census-region-by-year fixed effects.

The richer version adds weather, day-of-week/day-of-year controls, county income, and regulation-by-region trends:

$$\ln(y_{it}) = \alpha \cdot Treat_{ct} + \beta W_{it} + \gamma_r D_t + \delta I_{ct} + \theta Trend_{rct} + \mu_i + \eta_{ry} + \varepsilon_{it}.$$

Read: The coefficient vector α measures how ozone changes when a county is under a given gasoline standard, relative to a 9.0 psi RVP baseline, after absorbing monitor fixed effects and flexible regional time shocks.

5.2 Regression discontinuity design

The RD approach exploits the fact that gasoline content standards start at discrete dates. Because gasoline turns over quickly and ozone decomposes overnight, a regulation that changes emissions should produce a visible step change in ozone near the implementation date.

The monitor-specific time-series regression is:

$$\ln(y_{it}) = \alpha_i \cdot Treat_{ct} + \beta_i W_{it} + f_i(Date_t) + \mu_i + \varepsilon_{it},$$

where $f_i(Date_t)$ is a flexible eighth-order Chebychev polynomial in time for monitor i .

Interpretation: This specification lets every monitor have its own weather response, regulatory effect, and smooth time trend. Identification comes from abrupt changes at regulation start dates, not from long-run trends.

5.3 Why the two designs complement each other

- **DD strength:** uses the full national panel and compares treated to control counties over time.
- **DD weakness:** vulnerable if treated areas have nonlinear unobserved trends relative to control areas.
- **RD strength:** focuses on discontinuities around implementation dates and allows monitor-specific nonlinear trends.
- **RD weakness:** estimates can be noisy at individual monitors and depend on having a clear timing discontinuity.

Takeaway: The key claims are credible because both designs point in the same direction: federal RVP/RFG do not reduce ozone, while CARB does.

6 Main results

6.1 Federal RVP does not reduce ozone

The DD estimates show no economically or statistically significant reduction from RVP Phase II standards relative to the 9.0 psi baseline. The raw residual plots also show little discontinuity when the tight RVP standards begin.

Interpretation: Tightening vapor pressure alone is not enough. If refiners meet the standard by changing butane content, and if butane is weakly ozone-reactive, ozone concentrations need not fall.

6.2 Federal RFG shows mixed or weak effects

The DD specifications sometimes show a modest negative association between federal RFG and ozone, but the effect is sensitive to controls, trends, and monitor samples. The RD evidence suggests that apparent RFG reductions are concentrated in particular eastern locations and are more plausibly explained by contemporaneous NO_x control installations at electric generation plants.

Economic message: RFG may reduce some emissions, but the paper finds no robust evidence that it caused meaningful ozone reductions through its gasoline VOC component.

6.3 CARB substantially reduces ozone

CARB gasoline leads to large and statistically significant reductions in ozone, especially in southern California. The paper reports an average RD effect of about 15.6 percent in the Los Angeles–San Diego severe ozone nonattainment area, with reductions above 20 percent at some monitors.

Policy lesson: CARB is costly, but the benefits in high-ozone, densely populated California areas are large. A back-of-the-envelope calculation suggests that mortality benefits alone exceed costs.

6.4 Heterogeneity matters

The CARB effects are spatially heterogeneous. Inland southern California monitors show large reductions, while cooler or less densely populated counties show smaller or insignificant effects. This is consistent with ozone chemistry: CARB matters most where VOC reductions are binding and ozone levels are high.

7 Mechanism: refiners' behavioral response

7.1 Butane as the low-cost compliance margin

Federal RVP and RFG standards reduce the overall volume of VOC emissions but do not assign different weights to VOC compounds by ozone-forming reactivity. Refiners minimize costs by reducing butane, a volatile compound that raises gasoline's vapor pressure and is cheap to adjust.

Interpretation: This compliance choice satisfies the regulatory metric but does not target the chemical compounds most responsible for ozone formation.

7.2 Evidence from refinery production and ambient VOCs

Figure 10 shows strong seasonal butane production patterns after RVP and RFG rules begin: refiners produce butane in summer rather than blend it into gasoline, and then blend it back in winter. Figure 11 shows that ambient butane in Massachusetts dips in summer, unlike similar VOCs such as pentane and hexane.

Takeaway: The mechanism evidence supports the claim that federal rules changed gasoline composition, but in a way that was not environmentally effective for ozone.

7.3 Why CARB works differently

CARB imposes direct limits on highly reactive VOC classes, especially olefins and aromatics. This removes refiners' ability to comply solely by reducing butane. CARB is therefore a more restrictive rule but one that better aligns the regulated margin with ozone damages.

8 Cost-benefit and policy implications

- **Federal rules are costly despite weak ozone benefits.** RVP and RFG apply broadly in summer and segment fuel markets, raising consumer costs. The paper estimates that non-California federal VOC standards could raise annual gasoline expenditures by hundreds of millions of dollars.
- **CARB benefits appear large in the right places.** In California, the authors translate estimated ozone reductions into roughly 660 saved lives per year using mortality estimates from Bell et al. (2004). With CARB production costs of roughly 8–11 cents per gallon, the implied cost per life saved is below the EPA's value of a statistical life.
- **Targeting matters.** CARB should not necessarily be required everywhere. The ozone benefits are largest in high-ozone, VOC-sensitive, densely populated regions. Fine geographic targeting can, however, segment fuel markets further.
- **Flexible regulation is not automatically efficient.** Flexibility lowers costs conditional on delivering the right environmental margin. If firms comply by abating low-damage emissions, welfare may be lower than under a more restrictive targeted standard.

9 Potential limitations and interpretation cautions

- **Ozone is chemically complex.** VOC and NO_x interactions vary by geography, weather, and emissions mix. Estimated effects may differ over time as vehicle fleets, electric generation, and background emissions change.
- **RFG has other objectives.** The paper does not evaluate RFG's potential benefits for carbon monoxide or toxic carcinogenic pollutants. The conclusion concerns ozone reductions.
- **CARB statewide policy complicates targeting.** Because CARB is adopted statewide, the design cannot compare California treated and untreated counties after 1996. The RD approach instead relies on monitor-level discontinuities around March 1996.
- **Policy-cost calculations are simple.** The cost-benefit exercise is a back-of-the-envelope calculation focused mainly on mortality. A full analysis would include morbidity, environmental benefits, market segmentation, and spatial gasoline-market interactions.

10 Figures and tables

10.1 Figure 1: Regulatory Timeline and Figure 2: Map of RVP Phase II and RFG Regulations as of 2006

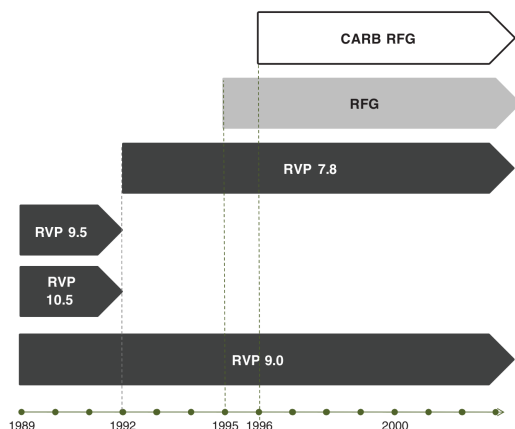


FIGURE 1. REGULATORY TIMELINE

: RVP: Reid vapor pressure regulation (the number refers to the vapor pressure limit). RFG: Federal reformulated gasoline. CARB RFG: California Air Resources Board reformulated gasoline.

Figure 1: Regulatory Timeline

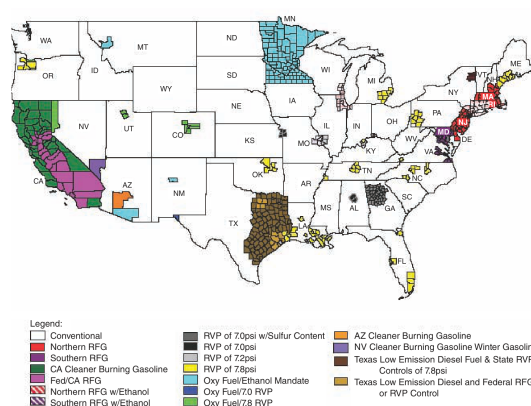


FIGURE 2. MAP OF RVP PHASE II AND RFG REGULATIONS AS OF 2006

Notes: Unshaded "conventional" gasoline areas are subject to the summertime RVP Phase II standard of 9.0 psi. Shaded areas in Minnesota, Colorado, Utah, and Montana have oxygenated gasoline for control of carbon monoxide pollution but do not have RVP or RFG. This study does not evaluate the effect of oxygenates on carbon monoxide pollution.

Source: EPA (Dec. 2006)

Figure 2: Map of RVP Phase II and RFG Regulations as of 2006

Read: Figure 1 summarizes the staggered introduction of RVP Phase I, RVP Phase II, federal RFG, and CARB gasoline. Figure 2 maps the geographic patchwork of fuel regulations as of 2006.

Interpretation: The figures make the identification strategy intuitive: gasoline regulations vary both over time and across space. They also show why gasoline regulation creates market segmentation: neighboring areas can face different fuel-content rules.

10.2 Table 1: Summary Statistics on Monitors and Regulation for the Summer Ozone Season and Figure 3: Mean Summer Ozone Concentrations

Table 1—SUMMARY STATISTICS ON MONITORS AND REGULATION FOR THE SUMMER OZONE SEASON (June–August)

Year	Observations/ (counties)	Counts of active monitors						
		Total monitors	Urban	Rural	RVPI ^a	RVPII ^b	RFG95	CARB
1989	63,076/(418)	720	153	244	371	0	0	0
1990	66,108/(436)	751	157	268	381	0	0	0
1991	69,164/(451)	782	151	297	395	0	0	0
1992	69,848/(452)	789	155	300	0	132	0	0
1993	72,606/(469)	815	167	301	0	140	0	0
1994	74,440/(473)	835	163	316	0	140	0	0
1995	77,007/(477)	865	170	330	0	111	111	0
1996	76,462/(471)	854	165	330	0	76	106	48
1997	78,283/(478)	873	166	336	0	76	108	48
1998	79,544/(487)	889	165	344	0	82	108	49
1999	80,750/(485)	899	168	344	0	87	108	49
2000	82,466/(489)	915	178	346	0	97	107	49
2001	83,781/(490)	929	178	355	0	97	108	47
2002	85,230/(495)	943	177	361	0	100	109	49
2003	85,260/(498)	945	180	362	0	101	108	50
Total	1,144,025/(NA)							
Average	76,268/(471)	854	166	322				

Notes: Data used in the construction of this table are monitor-days for which at least nine hours were recorded between 9 A.M. and 9 P.M., and monitor-years for which valid monitor-days were recorded for at least 75 percent of the days between June 1 and August 31. This table uses data for only the months June, July, and August, which constitute the main ozone season. Counties not counted in a RVPI, RVPII, RFG, or CARB column are subject to an RVP standard of 9.0 psi.

^aThe RVPI column lists, for 1989–1991, the number of counties with a 9.5 or 10.5 psi RVP requirement.

^bThe RVPII column lists, for 1992 onwards, the number of counties subject to an RVP of 7.8 psi or lower.

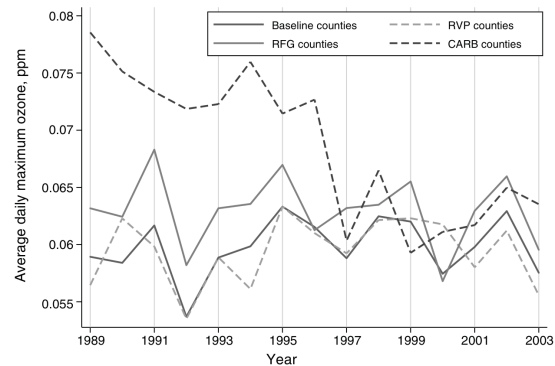


FIGURE 3. MEAN SUMMER OZONE CONCENTRATIONS
(Broken out by regulation type)

both Figures 3 and 4, data include only those monitors recording data in every summer.

Table 1: Summary Statistics on Monitors and Regulation for the Summer Ozone Season (June–August)

Figure 3: Mean Summer Ozone Concentrations

Read: Table 1 reports the number of monitor-days, monitors, counties, and treated counties by year. Figure 3 plots summer ozone averages by regulation type.

Economic message: The raw data show that treated counties tend to have higher ozone than baseline counties, especially CARB counties before treatment. This validates the need for monitor fixed effects and careful treatment timing rather than simple cross-sectional comparisons.

10.3 Figure 4: Mean Summer Daily Temperature Maxima and Figure 5: Mean Summer Ozone Concentrations - Panels A and B

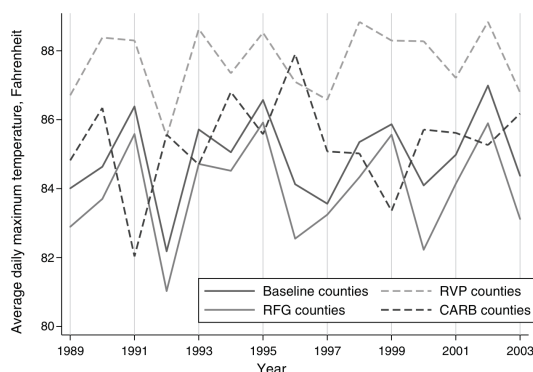


FIGURE 4. MEAN SUMMER DAILY TEMPERATURE MAXIMA
(Broken out by regulation type)

both Figures 3 and 4, data include only those monitors recording data in every summer.

Summer RVP of 9.5 nsi or 10.5 nsi (many counties 1989–1991)

Figure 4: Mean Summer Daily Temperature Maxima

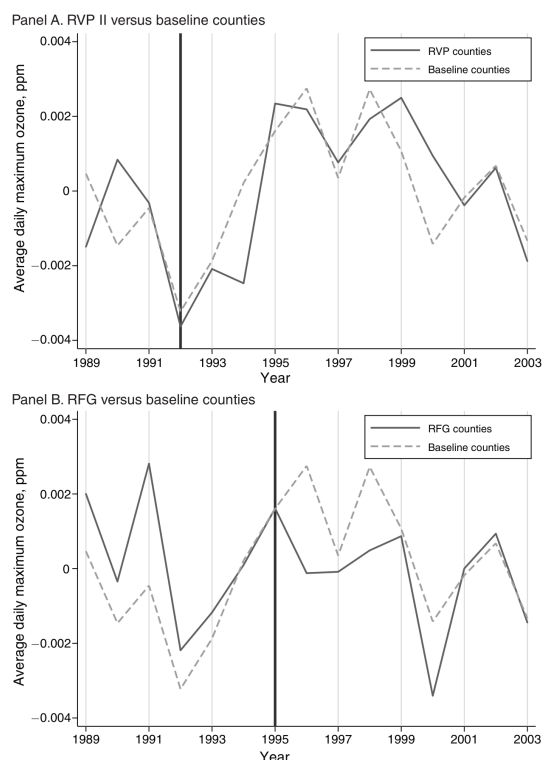


FIGURE 5. MEAN SUMMER OZONE CONCENTRATIONS

(Continued)

Figure 5: Mean Summer Ozone Concentrations - Panels A and B

Read: Figure 4 shows average summer maximum temperature by regulation type. Figure 5 panels A and B plot residualized ozone series for RVP II and RFG counties relative to baseline counties.

Interpretation: The close relationship between heat and ozone explains why weather controls are central. Figure 5 suggests no obvious RVP effect and only a modest possible RFG break, motivating the formal regressions.

10.4 Figure 5: Mean Summer Ozone Concentrations - Panel C and Table 2: Difference-in-Differences Estimation Results

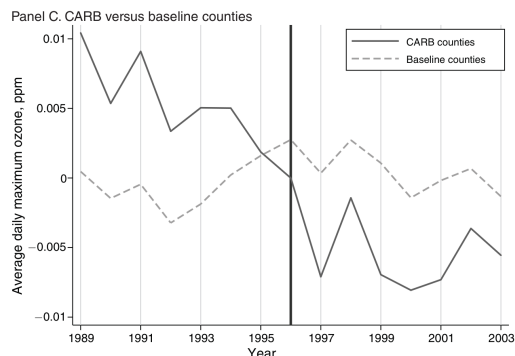


FIGURE 5. MEAN SUMMER OZONE CONCENTRATIONS (Concluded)

Values plotted are averaged residuals of a regression of daily maximum ozone on weather variables W_{it} described in Section IIIA and monitor fixed effects. Solid lines are TREATED counties; dashed lines are NLT counties. Vertical bars indicate the first implementation of the indicated regulation. Data include only monitors recording data in every summer.

Regressand	Dependent var: ln(daily maximum ozone concentration)					ln(daily max 8 hour concentration)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RVP Phase I:	0.016	0.013	0.015	0.001	0.004	0.018	0.015	0.004
9.5 or 10.5 psi	(0.016)	(0.015)	(0.016)	(0.016)	(0.018)	(0.017)	(0.017)	(0.020)
RVP Phase II:	-0.007	-0.011	-0.007	-0.012	-0.012	-0.005	-0.009	-0.011
7.8 psi or lower	(0.008)	(0.007)	(0.007)	(0.009)	(0.011)	(0.008)	(0.007)	(0.012)
Federal RFG	-0.029	-0.030	-0.016	-0.036	-0.019	-0.028	-0.028	-0.022
	(0.009)***	(0.007)***	(0.008)**	(0.011)***	(0.012)	(0.009)***	(0.008)***	(0.013)*
CARB gasoline	-0.095	-0.090	-0.077	-0.065	-0.064	-0.090	-0.086	-0.063
	(0.013)***	(0.011)***	(0.011)***	(0.019)***	(0.020)***	(0.013)***	(0.012)***	(0.021)***
County income (\$ billion)	—	—	-1.379	-0.225	-0.302	—	—	-0.102
	—	—	(0.285)***	(0.245)	(0.236)	—	—	(0.246)
Monitor FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region—year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region—DOW FEs	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Region FE—	No	Yes	Yes	Yes	Yes	No	Yes	Yes
DOY interaction								
Weather controls	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Income	No	No	Yes	Yes	Yes	No	No	Yes
Regulation—	No	No	No	Yes	Yes	No	No	Yes
region trends								
Regulation—	No	No	No	No	Yes	No	No	Yes
region quad trends								
Observations	1,144,025	1,144,025	1,144,025	1,144,025	1,144,025	1,144,025	1,144,025	1,144,025
R^2 (within-monitor)	0.024	0.261	0.261	0.264	0.264	0.026	0.255	0.259

Notes: Values shown are the coefficients of OLS regressions of the indicated dependent variable on the regressands. Standard errors clustered by state-year are in parentheses. All regulatory effects are relative to the omitted baseline of a 9.0 psi RVP standard. Sample uses data from June–August each year.

*** Significant at the 1 percent level.
 ** Significant at the 5 percent level.
 * Significant at the 10 percent level.

Figure 5: Mean Summer Ozone Concentrations - Panel C

Table 2: Difference-in-Differences Estimation Results

Read: Figure 5 panel C shows a sharp residualized decline for CARB counties after 1996. Table 2 reports DD estimates for RVP Phase I, RVP Phase II, federal RFG, and CARB relative to the baseline RVP 9.0 standard.

Takeaway: Table 2 is the main DD result. RVP estimates are small and insignificant. RFG estimates are modest and sensitive. CARB estimates are large, negative, and robust, implying a sizable ozone reduction.

10.5 Table 3: Difference-in-Difference Estimation Results: Urban versus Suburban versus Rural and Table 4: Difference-in-Differences Estimation Results: Monitors Recording Data in Every Year

TABLE 3—DIFFERENCE-IN-DIFFERENCE ESTIMATION RESULTS: URBAN VERSUS SUBURBAN VERSUS RURAL

Regressand	Dependent var: ln(daily maximum ozone concentration)					
	Urban		Suburban		Rural	
	(1)	(2)	(3)	(4)	(5)	(6)
RVP Phase I: 9.5 or 10.5 psi	0.020 (0.021)	0.007 (0.021)	0.024 (0.018)	-0.001 (0.016)	0.011 (0.018)	5.0E-04 (0.022)
RVP Phase II: 7.8 psi or lower	0.002 (0.014)	0.002 (0.014)	-0.011 (0.009)	-0.020 (0.011)*	-0.006 (0.009)	-0.011 (0.011)
Federal RFG	-0.004 (0.013)	-0.032 (0.015)**	-0.025 (0.009)***	-0.049 (0.014)***	-0.018 (0.011)	-0.027 (0.013)**
CARB gasoline	-0.071 (0.016)***	-0.072 (0.025)***	-0.113 (0.014)***	-0.102 (0.022)***	-0.040 (0.013)***	-0.017 (0.024)
County income (\$ billion)	-1.236 (0.397)***	0.470 (0.428)	-1.614 (0.268)***	-0.727 (0.219)***	-1.145 (0.629)*	0.051 (0.841)
Monitor FEs	Yes	Yes	Yes	Yes	Yes	Yes
Region—year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Region—DOW FEs	Yes	Yes	Yes	Yes	Yes	Yes
Region FE—DOY interaction	Yes	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes	Yes
Income	Yes	Yes	Yes	Yes	Yes	Yes
Regulation—region trends	No	Yes	No	Yes	No	Yes
Observations	222,982	222,982	490,539	490,539	430,504	430,504
R ² (within-monitor)	0.281	0.285	0.275	0.278	0.243	0.244

Notes: Values shown are the coefficients of OLS regressions of the indicated dependent variable on the regressands. Standard errors clustered by state-year are in parentheses. All regulatory effects are relative to the omitted baseline of a 9.0 psi RVP standard. Sample uses data from June through August each year.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 3: Difference-in-Difference Estimation Results: Urban versus Suburban versus Rural

TABLE 4—DIFFERENCE-IN-DIFFERENCES ESTIMATION RESULTS: MONITORS RECORDING DATA IN EVERY YEAR

Regressand	Dependent var: ln(daily maximum ozone concentration)					ln(daily max 8 hour concentration)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RVP Phase I: 9.5 or 10.5 psi	-0.009 (0.015)	-0.005 (0.015)	-0.005 (0.016)	-0.019 (0.016)	-0.014 (0.018)	-0.007 (0.016)	-0.003 (0.017)	-0.015 (0.020)
RVP Phase II: 7.8 psi or lower	-0.009 (0.009)	-0.014 (0.008)*	-0.009 (0.008)	-0.025 (0.010)**	-0.022 (0.012)*	-0.009 (0.009)	-0.013 (0.009)	-0.023 (0.013)*
Federal RFG	-0.031 (0.010)***	-0.034 (0.008)***	-0.018 (0.009)**	-0.055 (0.013)***	-0.031 (0.015)**	-0.031 (0.010)***	-0.034 (0.009)***	-0.036 (0.016)**
CARB gasoline	-0.148 (0.014)***	-0.144 (0.013)***	-0.118 (0.013)***	-0.134 (0.023)***	-0.123 (0.024)***	-0.139 (0.014)***	-0.137 (0.014)***	-0.121 (0.025)***
County income (\$ billion)	—	—	-1.851 (0.326)***	-0.289 (0.265)	-0.414 (0.247)*	—	—	-0.201 (0.270)
Monitor FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region—year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region—DOW FEs	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Region FE—DOY interaction	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Weather controls	No	Yes	Yes	Yes	Yes	No	Yes	Yes
Income	No	No	Yes	Yes	Yes	No	No	Yes
Regulation—region trends	No	No	No	Yes	Yes	No	No	Yes
Regulation—region quad trends	No	No	No	No	Yes	No	No	Yes
Observations	455,084	455,084	455,084	455,084	455,084	455,084	455,084	455,084
R ² (within-monitor)	0.031	0.278	0.280	0.285	0.285	0.030	0.270	0.280

Notes: Values shown are the coefficients of OLS regressions of the indicated dependent variable on the regressands. Standard errors clustered by state-year are in parentheses. All regulatory effects are relative to the omitted baseline of a 9.0 psi RVP standard. Sample uses data from June–August each year.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 4: Difference-in-Differences Estimation Results: Monitors Recording Data in Every Year

Read: Table 3 splits the DD estimates by monitor setting: urban, suburban, and rural. Table 4 repeats the DD analysis using only monitors with data in every year.

Interpretation: These robustness checks reinforce the central pattern. CARB remains large and negative. RVP remains ineffective. RFG continues to look mixed, and its estimates are sensitive to sample restrictions and trend controls.

10.6 Figure 6: Daily Maximum Ozone Concentrations - Panels A and B and Figure 6: Daily Maximum Ozone Concentrations - Panel C

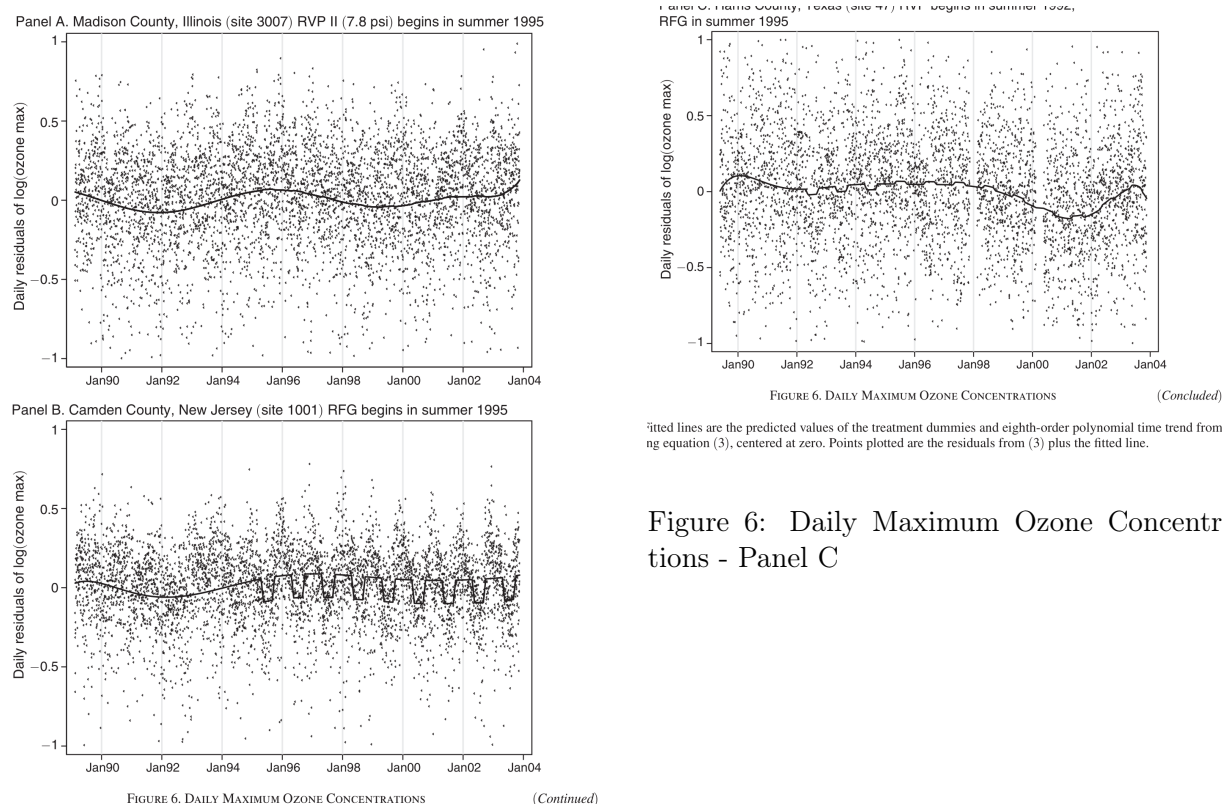


Figure 6: Daily Maximum Ozone Concentrations - Panel C

Figure 6: Daily Maximum Ozone Concentrations - Panels A and B

Read: Figure 6 illustrates the RD design for specific monitors. Panel A shows an RVP example in Madison County, Illinois; panel B an RFG example in Camden County, New Jersey; panel C a Harris County, Texas example with RVP and RFG.

Interpretation: These panels show why monitor-specific RD is useful. RVP and many RFG locations do not show convincing breaks. Some eastern RFG monitors show declines, but the paper later argues these are better explained by contemporaneous NO_x controls.

10.7 Figure 7: Distribution of Estimated RD Treatment Effects by Policy Type and Table 5: NO_x Control Installations at Electric Generation Plants by State and Year

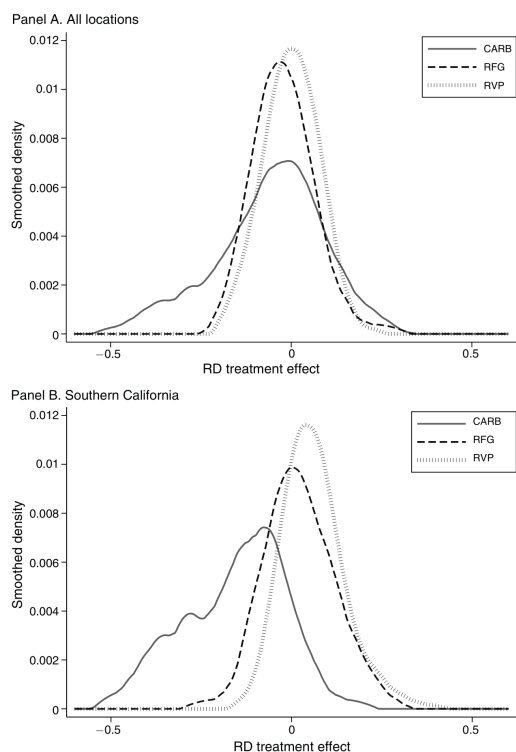


FIGURE 7. DISTRIBUTION OF ESTIMATED RD TREATMENT EFFECTS BY POLICY TYPE

The figures display the smoothed cross-monitor distribution of estimated treatment effects from equation smoother uses an Epanechnikov kernel with a bandwidth of 0.05. The RVP and RFG results shown in panel i include California. Panel B includes only Los Angeles, Orange, Riverside, San Bernardino, San Diego, and counties. These are the only counties in California in which RFG was ever implemented.

Figure 7: Distribution of Estimated RD Treatment Effects by Policy Type

Read: Figure 7 shows the cross-monitor distributions of RD treatment effects for CARB, RFG, and RVP. Table 5 records NO_x control installations at electric generation plants in key eastern states, California, and other states.

Economic message: The CARB distribution is shifted left, especially in southern California, indicating ozone reductions. The RVP and RFG distributions are centered near zero. Table 5 helps explain why some eastern RFG estimates look negative: NO_x controls were installed around the same time and location.

TABLE 5—NO_x CONTROL INSTALLATIONS AT ELECTRIC GENERATION PLANTS BY STATE AND YEAR

Year	Delaware, New Jersey, New York, Pennsylvania	California	All other states
1989	1	0	4
1990	5	0	7
1991	1	1	11
1992	7	0	15
1993	9	4	32
1994	33	6	60
1995	49	6	89
1996	19	0	47
1997	2	2	36
1998	8	0	60
1999	10	2	105
2000	5	4	68
2001	23	11	53
2002	54	4	68
2003	16	1	90

Source: Platts' BaseCase Continuous Emissions Monitoring System Unit Summary Data

Table 5: NO_x Control Installations at Electric Generation Plants by State and Year

10.8 Figure 8: Daily Maximum Ozone Concentrations and Figure 9: Maps of Estimated CARB RD Treatment Effects

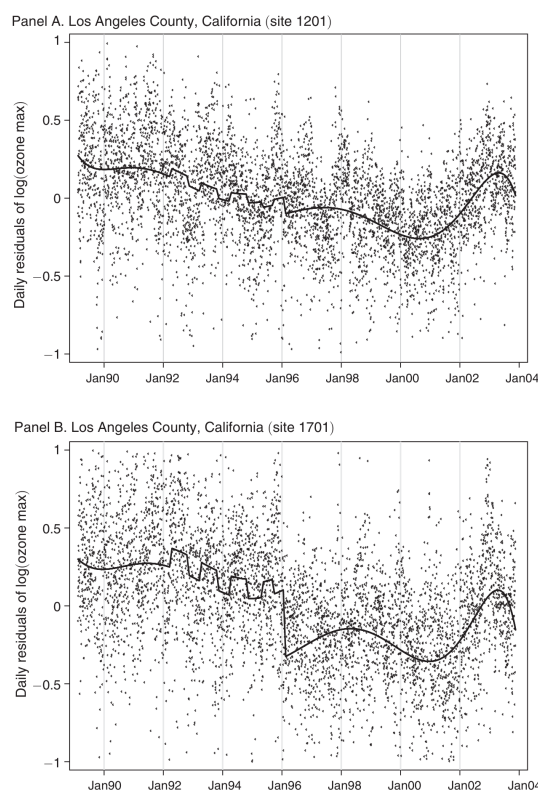


FIGURE 8. DAILY MAXIMUM OZONE CONCENTRATIONS

itted lines are the predicted values of the treatment dummies and eighth-order polynomial time trend from equation (3), centered at zero. Points plotted are the residuals from (3) plus the fitted line. Regulations RVP (summer 1992–1994), RFG (summer 1995), and CARB (March 1996–).

Figure 8: Daily Maximum Ozone Concentrations



FIGURE 9. MAPS OF ESTIMATED CARB RD TREATMENT EFFECTS

Figure 9: Maps of Estimated CARB RD Treatment Effects

Read: Figure 8 presents two Los Angeles County monitors with visible declines around CARB implementation. Figure 9 maps the spatial distribution of CARB RD estimates across California and southern California.

Interpretation: CARB's strongest effects are inland and in the Los Angeles–San Diego severe nonattainment region, where ozone formation is more likely to be responsive to reductions in reactive VOCs.

10.9 Figure 10: Refinery Net Production of Butane and Figure 11: Ambient VOC Concentrations in Massachusetts, 1992–1996

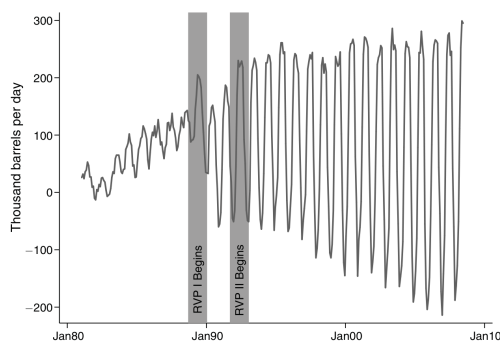


FIGURE 10. REFINERY NET PRODUCTION OF BUTANE

Notes: Positive values indicate that the quantity of butane refiners produce from crude oil exceeds the quantity blended into gasoline. Negative values indicate that refiners are blending more butane into gasoline than they produce from crude; that is, they are withdrawing butane from storage.

Figure 10: Refinery Net Production of Butane

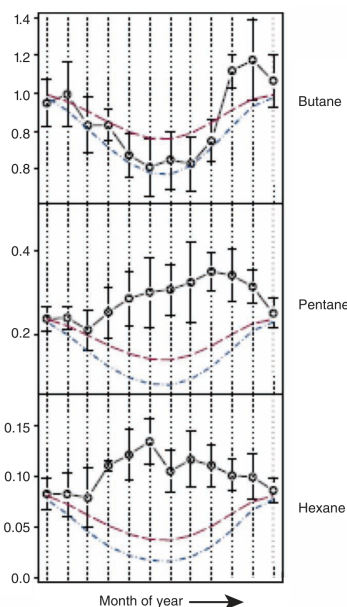


FIGURE 11. AMBIENT VOC CONCENTRATIONS IN MASSACHUSETTS, 1992–1996

Notes: The solid lines with markers represent the ratios of butane, pentane, and hexane concentrations relative to concentrations of acetylene. Relative measurements are used because acetylene is believed to be emitted at a constant rate across the year, and seasonal fluctuations in its concentration will be entirely due to atmospheric mixing effects that would similarly affect butane, pentane, and hexane.

Source: Reproduced by permission of American Geophysical Union.

Figure 11: Ambient VOC Concentrations in Massachusetts, 1992–1996

Read: Figure 10 shows that refinery net butane production becomes strongly seasonal after RVP and RFG rules begin. Figure 11 shows that ambient butane concentrations dip in summer in Massachusetts, unlike pentane and hexane.

Interpretation: These figures provide mechanism evidence. Federal rules changed gasoline composition, but largely by reducing butane in summer gasoline. Because butane is cheap to remove and weakly reactive in ozone formation, this explains why RVP/RFG compliance did not translate into ozone reductions.

11 Summary

- **Result.** Flexible federal gasoline standards reduced cheap-to-abate VOCs but did not reduce ozone; targeted California standards forced reductions in ozone-reactive VOCs and substantially improved air quality.
- **Identification.** Staggered county-level timing of gasoline standards plus monitor fixed effects, weather controls, region-year effects, regulatory trends, and monitor-specific RD around implementation dates.

- **Most important empirical table.** Table 2: RVP insignificant, RFG mixed/modest, CARB large and robustly negative.
- **Most important mechanism figures.** Figures 10 and 11: refiners reduced butane, and butane is not the ozone-relevant VOC margin.
- **Welfare intuition.** CARB is expensive but delivers large benefits in high-ozone California areas. RVP/RFG are cheaper per gallon but have little measured ozone benefit, so they can be socially wasteful for ozone control.
- **General lesson.** Flexibility is valuable only when the regulator defines the performance metric close to the true damage function. Otherwise firms will optimize the metric, not the environment.